Nestor Ngabonziza

AI Engineer · Full-Stack Developer

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PROFILE

I'm a software developer with over four years of experience building web and mobile apps, now focused on AI. I lead the team building Kumva's Farm Smarter platform, which turns sensor and farm data into clear advice for poultry farmers. In 2026 I took an intensive eleven-month AI specialization with MaibornWolff and Ojemba, and I'm studying for a Master's in Big Data Analytics. I'm passionate about building AI products people can trust: backed by real data, reliable on weak internet and available in the user's own language.

SKILLS

AI & LLMs	Prompt engineering · RAG · Embeddings · Vector databases · Tool calling · Structured outputs · LangChain · LangGraph · MCP · Multimodal AI · Evals and AI testing · Claude and OpenAI APIs
Backend & data	Python · FastAPI · Node.js · NestJS · Java (Spring Boot) · REST APIs · Microservices · PostgreSQL · MongoDB · Redis · InfluxDB · SQL · RabbitMQ
IoT	MQTT · Telegraf · Time-series data · Sensor data cleaning and calibration
Frontend & mobile	TypeScript · React · Next.js · React Native · Tailwind CSS · Material UI · Redux · MobX
Tools & teamwork	Docker · Kubernetes · Git · Strapi · Flutterwave · Agile / Scrum · Code review · Mentoring

EXPERIENCE

Team Lead, Kumva Farm Smarter Platform — Current

Kumva Insights · Kigali, Rwanda · kumva.io

Helps poultry farmers and Farm Extension Officers (FEOs) spot problems early and know what to do next.

- Lead the team building Farm Smarter: I plan the work, review code and make the main technical decisions across backend, data, AI and mobile.
- Designed the AI layer: Claude writes weekly reports and answers farmers' questions in Kinyarwanda, English and French. Every number must match the real data, and personal details never reach outside AI tools.
- Turned Kumva's poultry monitoring guide into daily rules: KPIs, a weekly brooding score from 0 to 100, and findings that keep facts apart from likely causes, each with a clear next step.
- Wrote the full backend design: 11 FastAPI services, each with its own PostgreSQL database, sharing events through RabbitMQ, with Redis for caching and live values.
- Mapped out the sensor data flow: temperature, humidity, CO_2 and ammonia readings come in over MQTT, Telegraf stores them in InfluxDB, and each one is checked, calibrated and matched to its house and flock.
- Defined alerts that follow the chicks' age (33 °C on day 0, down to 24 °C in week 3) and wait 30 minutes before firing, so farmers get the alerts that matter by push and SMS, not noise.
- Planned an offline-first React Native app: workers record dead birds, feed and water with no signal, and the app syncs once the phone is back online.

Tech: Python · FastAPI · PostgreSQL · Redis · RabbitMQ · MQTT · Telegraf · InfluxDB · React Native · Claude API · Kubernetes

Software Developer — Jun 2023 – Present

KB21 · Kigali, Rwanda

- Build web apps that load fast and are easy to use, with React, Next.js, TypeScript, Tailwind CSS and Material UI.
- Build REST APIs with Node.js and MongoDB, and mobile apps with React Native.
- Connect websites to Strapi so clients can update their own content.
- Mentor junior developers, review their code and help with testing before each release.
- Work in Agile sprints: planning, estimating and shipping in small, regular steps.

AI TRAINING & CERTIFICATIONS

AI Specialization: LLMs in Real-World Applications

Jan 2026 – Nov 2026

MaibornWolff and Ojemba

An intensive eleven-month program on building real products with large language models, with hands-on capstone projects along the way.

- Python and FastAPI: async APIs, databases, production-ready services
- LLM basics, prompt engineering and building a chat app
- AI system design: structured outputs, tool calling, conversation state, token and cost control
- RAG: document ingestion, chunking, embeddings, vector databases, similarity search
- Agents and workflows with LangChain and LangGraph
- Multimodal AI and the Model Context Protocol (MCP)
- Evaluating, testing and securing AI systems, plus error handling and reliability

Andela Technical Leadership Program

Nov 2021 – Oct 2022

Andela Rwanda · Kigali · technical and soft skills for a software engineering career

EARLIER EXPERIENCE

Software Developer Trainee

Sep 2022 – Jun 2023

The Gym · Kigali, Rwanda

- Finished an intensive full-stack training in JavaScript, React, Next.js and Tailwind CSS, building responsive, accessible web apps alone and in teams.
- Worked in Agile sprints and made code faster and easier to maintain through debugging and refactoring.

Frontend Developer Intern

Jul 2022 – Dec 2022

Justice Chatbot Ltd · Kampala, Uganda

- Helped build the interface of the OnDuka platform with React, Next.js and Tailwind CSS, making pages work well on phones and across browsers.
- Connected pages to APIs with the backend team, joined code reviews and kept a set of reusable components.

EDUCATION

Master of Science in Big Data Analytics

2025 – 2027 (in progress)

Adventist University of Central Africa

Bachelor of Science in Electronics and Telecommunication Engineering

2017 – 2022

University of Rwanda, College of Science and Technology

OTHER PROJECTS

- **Circulor** · circulor.com · Next.js, TypeScript, Tailwind CSS, Strapi
- **Saiz Dashboard** · saiz.io · Next.js, Tailwind CSS, Redux
- **Baumeister** · agentur-baumeister.com · Next.js, TypeScript, Tailwind CSS
- **Co-Design** · enaccess.github.io/co-design-FE · Next.js, TypeScript, Tailwind CSS
- **CSC SensorHub** · Google Play · React Native, MobX, TypeScript

LANGUAGES

Kinyarwanda – native | **English** – fluent | **French** – basic